

GRAVITYSORT · DESIGN DECISION RECORD

GravitySort recommendations accepted

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DOCUMENT CONTROL

Document ID	GVS-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/decisions/0002-recommendations-accepted.md @ 1e09290		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of the recommendations in GVS-DDR-001 and docs/REVIEW.md, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item in GVS-DDR-001 and `docs/REVIEW.md` that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items with no recommendation remain proposed, awaiting Amish.

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." GVS-DDR-001 (v0.1) had adopted most TRL 2 recommendations for TRL 3 work, open for his review, and listed the budget and six TRL 3 items as awaiting him. This record turns every item that had a recommendation into a decision, applies it at TRL 3 level inside this repo, and lists what is still open. Where a recommendation offered several options, the recommended option is the decision. TRL 4 work (building, testing, purchasing) stays on hold by Amish's instruction, and the project stays at TRL 3.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

# (GVS-DDR-001)	Decision	What changed in the repo
1	Budget raised from \$350 to \$450	<code>project.yaml</code> <code>budget_usd</code> 350 to 450; R9 target \$450 (GVS-REQ-001 v0.4); README budget line; GVS-CAL-001 v0.2 section 12. The BOM is \$455, so R9 is still not met, by \$5
2	Two stages: centrifugal bowl, then shaking table	Status wording only (GVS-PRC-001 v0.4, GVS-DDR-001 v0.2)
3	Fluidized bowl with a rotary union; non-fluidized bowl kept as a documented low-cost variant	Status wording only
4	Cast polyurethane liner in a 3D-printed mold on a GFRP shell	Status wording only
5	Pedal crank baseline, MotionCore 250 W option, one drive time-shared between bowl and table	Status wording only
6	Battery for the motor option left to the user; SwapCell not named	Status wording only
7	Direct smelting with borax as the downstream step, outside this hardware	Status wording in GVS-PRB-001 v0.4
8a	Partner screening criterion: a country with a mercury ban in force (Colombia) or a Minamata action plan with a planetGOLD programme	Status wording in GVS-PRB-001 v0.4
9	R12 restated (option a): with the pedals, a burst safety factor of 10 or more at the highest reachable speed (1,200 rpm at a 100 rpm cadence) plus the speed display replaces the gearing cap; the motor stays capped at 900 rpm	GVS-REQ-001 v0.4 R12; GVS-CAL-001 v0.2 sections 10 and 14 and <code>sizing.py</code> . R12 moves from not met as written to met on paper (lowest factor 17)
10	Lighter frame (option b): 25 x 25 x 1.5 mm tube for the base frame, pedal outrigger, table stretchers and table head post, then re-estimate	<code>cad/src/model.py</code> (<code>tube</code> 30 to 25, new <code>tube_wall</code> 1.5; cross members and stand stretchers follow the tube size); STEP and STL re-exported; GVS-DWG-001 Rev P1 to P2; BOM items 1 (\$40 to \$32) and 15 (\$30 to \$28); GVS-CAL-001 v0.2 adds a frame member check (61 MPa, safety factor 3.9, 0.73 mm deflection). Mass 88.3 kg to 77.3 kg , heaviest load 25.8 to 18.1 kg: R11 met on paper
11a	Fluidization supply: 3/4 in hose and about 89 holes of 1.0 mm graded toward the lower rings	Already in the model notes and BOM; status wording in GVS-PRC-001 v0.4 and GVS-CAL-001 v0.2 section 6
14	Engineering proposals: disc brake with parking latch and lid interlock pin (BOM item 18); bicycle computer speed display (item 19); 1.2:1 motor chain step-up; MotionCore lid switch on a brake input	Already in the model, BOM and calculation; status wording in GVS-PRC-001 v0.4. The MotionCore assumptions are listed as cross-repo actions in <code>docs/REVIEW.md</code>

No pitch or problem rewording was recommended, so the `project.yaml` pitch and problem are unchanged.

Items still open

Table 2. Proposed, awaiting Amish (no recommendation was made).

# (GVS-DDR-001)	Item	Why it is open
1 (part)	Cost-down options to close the \$5 gap: quarter-turn belt instead of the bevel gearbox (about \$25), non-fluidized variant (about \$45), table later (about \$70 deferred)	Listed as options with no recommendation
8b	The first co-design partner itself	No partner was recommended; only the screening criterion
11b	Head for the fluidization supply: a 1.7 m post or a small pump, for 10 kPa net at every ring	No preference was stated between the two
12	Water pumping from the settling pond at pedal-only sites (about 1.19 m ³ /h, 5.5 W hydraulic)	No recommendation yet
13	R2 and R5 margins: raise the design feed to about 210 kg/h, or restate R2 as 200 kg/h of feed time	Two options, no recommendation

On hold (TRL 4)

Nothing decided here needs TRL 4 work to be recorded. The checks the decisions point at, such as bearing alignment on the first 25 mm frame, spin-up with the tub and lid as containment, and brake stop time, are TRL 4 work and stay on hold by Amish's instruction.

■ Consequences

- GVS-PRB-001, GVS-PRC-001 and GVS-REQ-001 move to v0.4, GVS-CAL-001 to v0.2 and GVS-DDR-001 to v0.2, each with a revision entry for this record.
- Requirements now stand at one not met (R9, \$455 against \$450), three at risk (R2, R5, R6), two not verifiable before testing (R4, R14) and eight met on paper (R1, R3, R7, R8, R10, R11, R12, R13).
- The pedal drive is still not speed-capped; the restated R12 makes the structural margin and the speed display carry that risk, and the tub and lid are not yet shown to contain a liner fragment.
- `trl` and `trl_target` stay at 3.